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School of Electronic and Electrical Engineering

ELEC5870M Final Report

Fabrication of Enhancement Mode HEMT Devices via Gate Recession

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Abstract

Abstract

This report documents the complete ELEC5870M MEng project on fabricating enhancement-mode GaAs/AlGaAs high-electron-mobility transistors via controlled gate recession. It will first summarise the validated baseline process and depletion-mode electrical benchmarks established during Semester 1 before detailing the Semester 2 process innovations that enabled tightly-controlled recess depths. The final chapters will correlate recess depth with threshold voltage, transconductance, and gate leakage to demonstrate the tuning range achieved, and will reflect on the manufacturability of the proposed flow.

Acknowledgements

I would like to thank Dr Christopher Wood for supervision throughout this project and Mr Rob Farr for his consistent cleanroom guidance. I am also grateful to the Leeds Nanotechnology cleanroom staff, including Mark Rosamond, Rory Gilroy, and Thomas Moorhouse, for their expertise during training sessions and to Professor Edmund Linfield for insightful discussions on III–V device physics.

Abbreviations

2DEG	Two-dimensional electron gas
B1500A	Semiconductor parameter analyser
DI	Deionised water
HEMT	High-electron-mobility transistor
HCl	Hydrochloric acid
III-V	Group III-V semiconductor
IPA	Isopropanol
LOR	Lift-off resist
MLA	Maskless aligner
MMIC	Monolithic microwave integrated circuit
RTA	Rapid thermal annealing
SEM	Scanning electron microscopy

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CHAPTER 1

Introduction

Modern communication, sensing, and microwave systems impose increasingly demanding requirements on the semiconductor devices that form their front-end signal paths. Higher operating frequencies, tighter noise constraints, and strict linearity targets continue to push transistor technologies beyond the capabilities of many silicon-based platforms. [1] Although silicon processes have made notable progress in recent years, they often fall short in providing the necessary electron mobility, noise performance, or high-frequency gain required for performance-critical microwave and millimetre-wave applications (e.g. silicon's $1200 \text{ cm}^2/\text{Vs}$ compared to GaAs's $6500 \text{ cm}^2/\text{Vs}$ in transistors for MMICs). [1]

III–V High–Electron–Mobility Transistors (HEMTs) play a central role in addressing these challenges. Their heterostructure-based design creates a highly conductive channel at the interface between two semiconductor layers with different bandgaps. At this interface a two-dimensional electron gas (2DEG) forms, supporting exceptionally high carrier mobility. The resulting strong transconductance, stable high-frequency behaviour, and low-noise performance have firmly established HEMTs as core devices throughout microwave and millimetre-wave circuit design. [2]

Research activity within the field has increasingly shifted toward gallium nitride (GaN) HEMTs, driven by the material properties enabled by its wide bandgap. GaN supports high breakdown voltages, strong power-handling capability, and robust operation under large electric fields, making it the preferred choice for many high-power and high-linearity applications. [3] Despite this growing dominance, gallium arsenide (GaAs) HEMTs remain one of the most mature and well-characterised III–V transistor platforms. Their stability, extensive documentation, and long industrial history continue to make them an invaluable system for studying the relationships between heterostructure design, device architecture, and electrical behaviour. [4, 5] For the purposes of this project, GaAs presents clear practical and technical advantages. The Leeds Nanotechnology Cleanroom maintains established, reliable processes for GaAs/AlGaAs heterostructures, and suitable wafers are readily available with reproducible epitaxial quality. [6]

Standard GaAs/AlGaAs HEMTs are inherently depletion-mode devices (d-mode); they naturally conduct current at zero gate bias and require a negative voltage to switch off. [5] This ‘normally-on’ behaviour is often undesirable as it necessitates complex negative-polarity power supplies [7] and complicates the design of direct-coupled logic circuits. [8] Consequently, there is significant interest in developing enhancement-mode devices (e-mode) that are ‘normally-off’ and offer reduced power consumption.

One common approach to achieving e-mode operation is to recess the gate into the AlGaAs donor layer, thereby reducing the charge carriers available to form the channel and shifting the

threshold voltage in the positive direction. [9] Achieving e-mode operation through gate recessing is inherently non-trivial: it requires precise and repeatable removal of only a few nanometres of material, with small variations in etch depth leading to significant changes in threshold voltage and device behaviour. [10]

The primary objective of this project is to investigate the relationship between gate recess depth and threshold voltage in GaAs/AlGaAs HEMTs, with the aim of achieving enhancement-mode operation, and with a focus on understanding the practical limitations and repeatability of the fabrication process. This was investigated through the development and iterative refinement of a gate recession process in which the gate contact is recessed into the AlGaAs donor layer to shift the threshold voltage into the positive region.

The work proceeds in two phases. The first phase established a validated depletion-mode baseline: a full set of unrecessed GaAs/AlGaAs HEMT devices was fabricated and electrically characterised, confirming that threshold voltage is governed by the vertical heterostructure and is independent of lateral device geometry. This baseline provides the reference against which all subsequent recessed devices are compared. The second phase developed a controlled gate recess etch process through three successive device fabrication iterations, each designed to diagnose and address specific failure modes identified in the previous run, including bypass current paths, mask design limitations, over-etching, and gate-semiconductor interface degradation.

Report Organisation

Chapter 2 reviews the device physics of GaAs/AlGaAs HEMTs and establishes the analytical relationship between donor-layer thickness, the quantum-mechanical correction to the gate-to-channel separation, and threshold voltage. This framework is then used to predict the critical recess depth required for enhancement-mode operation in the specific heterostructure employed in this work. Chapters ?? and 4 document the baseline depletion-mode fabrication process and its electrical characterisation respectively, establishing the reference dataset against which all recessed devices are compared. Chapter 5 develops and calibrates the gate recess etch process, including its chemistry, oxide-mitigation strategy, and the mask architecture used to sweep recess depth across multiple devices. Chapters 6, 7, and 8 each present a successive fabrication iteration; each was motivated by a specific failure mode identified in its predecessor, and together they map the practical boundaries of the process. Chapter 9 consolidates these outcomes into a broader failure analysis and discussion of process limitations, and Chapter 10 draws conclusions and recommends directions for future work.

CHAPTER 2

Technical Background

2.1 Overview of GaAs/AlGaAs HEMTs

In GaAs/AlGaAs high-electron-mobility transistors (HEMTs), the interface between the GaAs channel layer and the overlying AlGaAs barrier produces a discontinuity in the conduction band due to the differing bandgaps of the two materials. This offset creates a potential well on the GaAs side of the junction, as illustrated in Figure 2.1, which confines electrons to the heterointerface and enables the formation of a two-dimensional electron gas (2DEG).

Silicon donors in the AlGaAs layer ionise and release electrons, which transfer across the heterointerface and accumulate in the potential well on the GaAs side. This modulation-doping scheme leaves the ionised donors in the AlGaAs while the mobile carriers reside in the undoped GaAs channel, spatially separating the two. The resulting suppression of ionised-impurity scattering yields the exceptionally high electron mobility, high cut-off frequencies, low noise, and strong transconductance that distinguish HEMTs from bulk-channel field-effect transistors. [11]

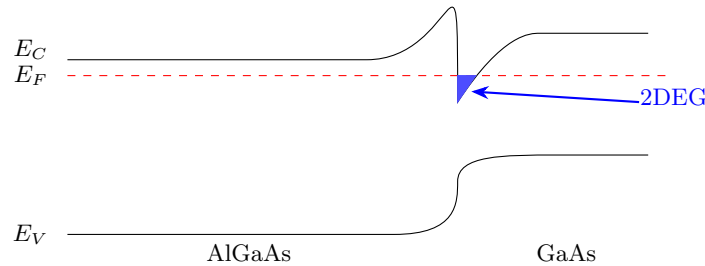


Figure 2.1: Band Structure Diagram of the AlGaAs/GaAs interface demonstrating the resulting Two-Dimensional Electron Gas (2DEG) formed in a Potential Well.

2.2 The GaAs/AlGaAs Heterostructure

The devices investigated in this project employ a conventional GaAs/AlGaAs heterostructure in which a sequence of epitaxial layers is engineered to create and control a 2DEG. A schematic cross-section of the structure is shown in Figure 2.2, highlighting the cap, donor, spacer, and channel layers whose thicknesses and compositions govern the electrostatic control and transport behaviour of the device.

At the surface of the structure lies a 10 nm thick, highly doped GaAs cap layer. This cap provides a chemically stable surface for forming the Schottky gate contact whilst protecting the underlying AlGaAs from oxidation. Its high conductivity reduces series resistance in the access regions adjacent to the source and drain contacts, improving carrier injection into the channel.

Beneath the cap layer is a 40 nm thick silicon-doped AlGaAs donor layer. Silicon donors

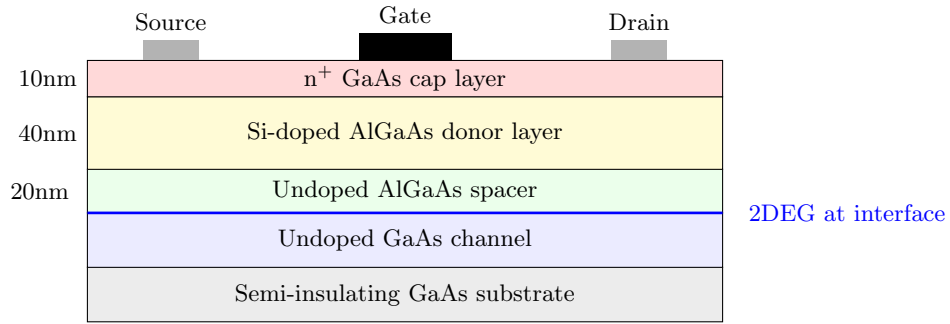


Figure 2.2: Cross-section of the GaAs/AlGaAs HEMT used in this project (not to scale).

act as electron suppliers, with carriers transferring into the GaAs channel as a result of the conduction band discontinuity at the heterointerface. This modulation-doping strategy spatially separates the ionised dopants from the conducting channel. Both the donor concentration and its thickness strongly influence the resulting sheet carrier density of the 2DEG and therefore the threshold voltage and maximum drain current of the device.

A 20 nm thick undoped AlGaAs spacer layer separates the donor layer from the GaAs channel. This spacer suppresses Coulomb scattering by physically isolating the 2DEG from ionised impurities, enabling exceptionally high electron mobility. [12] The benefit is particularly pronounced in HEMT devices operating at cryogenic temperatures, where phonon scattering is reduced and impurity scattering would otherwise dominate transport behaviour. [13] Increasing the spacer thickness improves mobility but weakens electrostatic coupling between the gate and the channel; this trade-off is a fundamental consideration in heterostructure design.

In this project, the thickness of the doped AlGaAs donor layer beneath the gate represents the primary controllable parameter through which the threshold voltage may be engineered.

2.3 The Recessed-Gate Architecture

In a typical HEMT device shown in Figure 2.2, the gate is deposited directly onto the GaAs cap with the full donor layer intact beneath it. The resulting threshold voltage is strongly negative, placing the device in depletion mode. A method of realising enhancement-mode operation, is to thin the donor layer by etching away the cap and penetrating a controlled depth into the AlGaAs, a process known as gate recession.

Figure 2.3 shows the resulting cross-section. The etch opens a trench through the cap and into the donor layer, into which the Schottky gate is deposited; the gate metal coats the walls of the trench and rests on the cap surface either side of the trench, whilst the source and drain contacts remain on the original surface, preserving low-resistance access to the channel. The thickness of donor remaining beneath the gate, d_d , is the critical process parameter: once it is reduced below a threshold value, the zero-bias depletion region extends to the 2DEG and the device switches to enhancement mode.

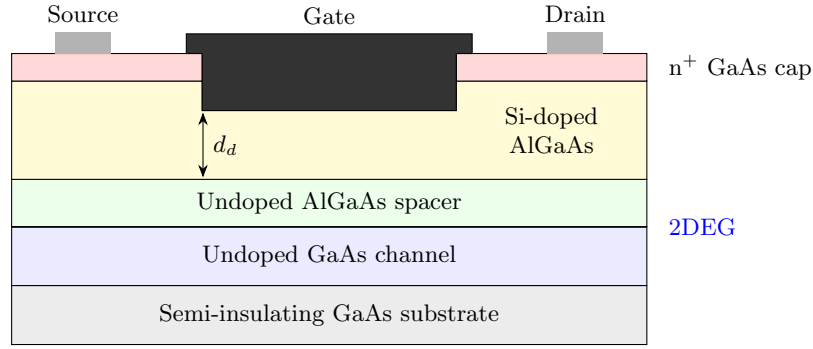


Figure 2.3: Cross-section of the recessed-gate GaAs/AlGaAs HEMT (not to scale).

2.4 Gate Control of Channel Charge and Threshold Voltage

The electrical behaviour of the GaAs/AlGaAs HEMT is governed by the field-effect interaction between the Schottky gate and the 2DEG. When a metal gate is deposited onto the semiconductor surface, a depletion region forms beneath it; the depth of this region is modulated by the applied gate voltage (V_G). This modulation controls the sheet carrier density (n_s) in the channel and, consequently, the drain current. [14]

2.4.1 Threshold Voltage and Required Recess Depth

The distinction between depletion-mode and enhancement-mode operation is governed by the threshold voltage V_{off} . Within the analytical framework of Lee *et al.* [15]:

$$V_{off} = \phi_b - \Delta E_c - \frac{qN_d d_d^2}{2\varepsilon} \quad (2.1)$$

ϕ_b : Schottky barrier height; ΔE_c : conduction band offset; N_d : donor concentration; d_d : donor layer thickness; ε : permittivity of the AlGaAs barrier.

In the unrecessed device, $d_d = 40$ nm and the quadratic term in Equation 2.1 dominates, yielding negative V_{off} values; the device is normally on and requires a negative gate bias to deplete the channel. [5] Enhancement-mode operation requires d_d to be reduced until V_{off} becomes positive.

A critical donor layer thickness is derived by setting $V_{off} = 0$ and solving for d_d in Equation 2.1.

$$d_{d,crit} = \sqrt{\frac{2\varepsilon(\phi_b - \Delta E_c)}{qN_d}} \quad (2.2)$$

$$d_{d,crit} = \sqrt{\frac{2 \times 12.2 \times 8.854 \times 10^{-12} \times (0.80 - 0.23)}{1.6 \times 10^{-19} \times (1 \times 10^{18}) \times 10^6}} = \sqrt{7.70 \times 10^{-16}} \approx 28 \text{ nm} \quad (2.3)$$

This is consistent with the ~ 25 nm quoted by Lee *et al.* [15] for a similar heterostructure, with the small difference attributable to the parameter values used. The required total etch

Parameter	Symbol	Value	Source
Schottky barrier height	ϕ_b	0.80 V	[15]
Conduction band offset	ΔE_c	0.23 eV	[11]
Relative permittivity	ε_r	12.2	[11]
Vacuum permittivity	ε_0	$8.854 \times 10^{-12} \text{ F m}^{-1}$	Physical constant
Donor concentration	N_d	$1 \times 10^{18} \text{ cm}^{-3}$	Wafer specification

Table 2.1: Material parameters used to evaluate the critical donor thickness for the $\text{Al}_{0.3}\text{Ga}_{0.7}\text{As}/\text{GaAs}$ heterostructure.

depth is:

$$\text{Recess Depth} = d_{\text{cap}} + (d_{\text{donor}} - d_{d,\text{crit}}) = 10 + (40 - 28) = 22 \text{ nm} \quad (2.4)$$

The value of 22 nm serves as a theoretical reference rather than a precise process target. In this project, multiple device sets are fabricated with recess depths spanning approximately 10–30 nm, sweeping through the predicted threshold condition. Devices etched below 22 nm are expected to exhibit progressively less negative V_{off} as d_d decreases; those etched beyond 22 nm should cross into enhancement mode. This systematic variation allows the dependence of threshold voltage on recess depth to be characterised empirically, as well as any other affected characteristics such as transconductance and drain current. As Equation 2.2 is evaluated using literature parameter values rather than direct measurement on these wafers, measurements of fabricated devices will enable the validation of the theoretical prediction and identify the true threshold depth for this specific heterostructure.

2.4.2 Charge Distribution and Effective Gate Separation

The threshold calculation above treats the 2DEG as a sheet of charge located exactly at the GaAs/AlGaAs heterointerface. In the charge-control model, the gate metal and 2DEG are treated as the two plates of a parallel-plate capacitor, separated by the total gate-to-channel distance d ; the capacitive coupling arises through the depleted AlGaAs barrier stack, with the gate voltage modulating the extent of the depletion region and the model assigning a uniform permittivity ε to this region. [15]

In practice, electrons forming the 2DEG occupy a quantised triangular potential well on the GaAs side of the heterointerface; the charge distribution has finite thickness and its centroid is displaced into the GaAs. Lee *et al.* account for this by introducing an effective gate-to-channel separation of $d + \Delta d$, where $\Delta d \approx 80 \text{ \AA}$ is a quantum-mechanical correction associated with the spatial extent of the electron wavefunction. [15] This correction enters the transconductance parameter β :

$$\beta = \frac{\varepsilon \mu W}{(d + \Delta d)L} \quad (2.5)$$

ε : Permittivity of the barrier; μ : Electron mobility; W , L : Gate width and length; d : Physical gate-to-channel separation; Δd : Effective charge-centroid offset.

The physical separation d comprises the residual donor layer and the spacer. For the unre-

cessed device, d spans the full epitaxial stack above the 2DEG:

$$d_{\text{unrecessed}} = d_{\text{cap}} + d_{\text{donor}} + d_{\text{spacer}} = 10 + 40 + 20 = 70 \text{ nm}$$

After recessing to the threshold condition (Equation 2.3), the cap is fully removed, the donor is thinned to $d_{d,\text{crit}} = 28 \text{ nm}$, and the spacer remains intact:

$$d_{\text{recessed}} = d_{d,\text{crit}} + d_{\text{spacer}} = 28 + 20 = 48 \text{ nm}$$

The significance of Δd in each configuration is therefore:

- **Unrecessed Gate:** $d_{\text{eff}} = 70 + 8 = 78 \text{ nm}$; the correction represents $8/78 \approx 10\%$ of the effective separation.
- **Recessed Gate:** $d_{\text{eff}} = 48 + 8 = 56 \text{ nm}$; the correction represents $8/56 \approx 14\%$ of the effective separation.

The relevance of Δd to this project is its effect on β . The correction arises from quantum confinement at the heterointerface: 2DEG electrons do not sit precisely at the interface but are distributed into the GaAs in a triangular potential well, so the effective charge centroid lies $\approx 8 \text{ nm}$ deeper than the physical boundary. This offset is independent of gate position — recessing the gate brings the metal closer to the interface, but it does not alter where the 2DEG charge sits. As d decreases with etching, Δd therefore represents a growing share of d_{eff} : a minor 10% correction in the unrecessed device, rising to 14% at the threshold recess depth. A model using physical geometry alone would overestimate β and transconductance by this margin, so the correction must be included when comparing measured device characteristics against theoretical predictions.

CHAPTER 3

Fabrication of Depletion-Mode HEMT Devices

3.1 Overview of the Fabrication Workflow

The depletion-mode HEMTs characterised in this report were fabricated using the standard III-V device process established within the Leeds Nanotechnology Cleanroom. This workflow follows the routine sequence used for GaAs/AlGaAs transistors in this facility: it consists of 3 main stages: mesa definition, ohmic contact formation, and gate formation. This well-established procedure was followed without modification and carried out over four full-day laboratory sessions. The devices produced through this workflow provide a necessary baseline; they allow for a reliable comparison against which the behaviour of e-mode devices with different structures and dimensions can be evaluated later in this project.

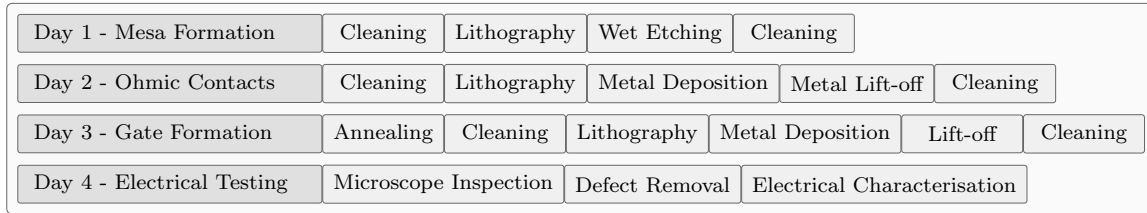


Figure 3.1: Timeline overview of the four-day cleanroom workflow used to fabricate and characterise baseline depletion-mode GaAs/AlGaAs HEMTs.

3.2 Mesa Definition and Isolation Etch

The fabrication process began with the electrical isolation of individual device regions through mesa etching. Prior to lithography, the GaAs wafer was cleaned sequentially in acetone, isopropanol (IPA), and deionised water (DI) to remove organic residues and ensure sufficient photoresist adhesion. To further enhance the bonding interface between the substrate and the resist, a Surpass primer was spun onto the wafer. This priming step is critical for wet-etching applications: it modifies the surface tension to prevent the etchant from penetrating under the resist edges which would otherwise lead to the delamination of the mask. A positive photoresist (S1803) was then spun onto the sample at 3000 rpm, soft-baked on a hot-plate at 115 °C, and subsequently patterned with exposure from the Heidelberg MLA 150 Maskless Aligner (MLA) [16].

Following exposure, the sample was immersed in solution of MF-351 developer diluted with DI (Ratio 1 : 3.5) for 45 seconds, followed by a thorough two-stage deionised (DI) water rinse and nitrogen blow-dry to reveal the mesa patterns. The mesa pattern was then transferred into the wafer using a wet-etch solution of sulphuric acid, hydrogen peroxide, and water

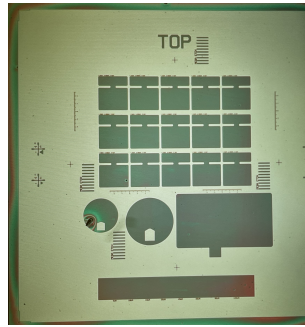


Figure 3.2: Mesa lithography mask used for defining isolated device regions (Post-Development).

($H_2SO_4:H_2O_2:H_2O$ in a 1:8:160 ratio).

This etchant is well-suited for GaAs/AlGaAs mesa formation, as it provides a controllable, moderately fast isotropic etch that removes the active layers uniformly [17]. The etch rate is slow enough for precise timing of a shallow ≈ 200 nm recess, yet fast enough for routine device isolation.

To calibrate the process, an initial 30-second pilot etch was performed. Subsequent measurement with an Alpha-Step profiler showed an etch depth of 114 nm, indicating an initial rate of 3.8 nm/s. To achieve the target depth of 200 nm, an additional 86 nm of material needed removal. Assuming a constant etch rate, a second etch of 22 seconds was calculated and carried out. However, the final measured depth was only 177 nm, meaning the second etch removed just 63 nm at a reduced effective rate of approximately 2.9 nm/s. Although 177 nm is adequate for device isolation, successfully penetrating the active GaAs/AlGaAs heterostructure layers, the rate discrepancy underscores the sensitivity of the wet-etch chemistry.

3.2.1 Analysis of Etch Rate Discrepancy

The main hypothesis for the reduced average etch rate in the second step is the formation of a native oxide layer on the GaAs surface during the interval when the sample was removed for profilometry. This oxide creates an “incubation period” at the start of the second etch, delaying material removal while the etchant penetrates the oxide to reach the underlying semiconductor. Since the second etch (22 seconds) was shorter than the first (30 seconds), this non-productive period accounted for a larger fraction of the total time, lowering the overall average rate.

A more specific attribution points to rapid oxidation of the AlGaAs donor layer surface during the profilometry interval. Unlike pure GaAs, which forms relatively unstable oxides, AlGaAs reacts with atmospheric oxygen to create a robust native oxide layer primarily composed of thermodynamically stable aluminium oxide (Al_2O_3), along with Ga_2O_3 and As_2O_3 [18, 19].

The $H_2SO_4:H_2O_2:H_2O$ wet-etch mechanism involves a continuous cycle: surface oxidation by the peroxide, followed by acid dissolution of the resulting oxide [17]. However, the air-grown, Al-rich passivation layer dissolves much more slowly than the transient, Al-poor oxides formed in situ during steady-state etching [19]. As a result, the second etch step includes a substantial incubation period to breach this stable Al_2O_3 cap before resuming the intrinsic etch rate of 3.8 nm/s. In a brief 22-second etch, this nonlinear delay constitutes a significant portion of the process, reducing the calculated average rate.

3.3 Ohmic Contact Formation

The formation of low-resistance Ohmic contacts is a critical prerequisite for accessing the two-dimensional electron gas (2DEG) within the heterostructure. When metal is deposited directly onto the GaAs cap, the interface naturally forms a Schottky barrier due to Fermi-level pinning at the surface states: this creates a rectifying junction that restricts carrier injection. To overcome this, a specific metallisation scheme and thermal processing step are required to facilitate field-emission (tunnelling) conduction. The objective is to create a heavily doped n^+ region beneath the contact pads that extends down towards the channel; this high doping density significantly narrows the depletion width of the barrier, allowing electrons to tunnel freely between the metal and the semiconductor with linear $I - V$ characteristics.

To define these contacts, a dual-layer photoresist stack comprising Lift-Off-Resist (LOR) beneath S1803 was employed. A single photoresist layer typically develops near-vertical sidewalls: this causes the evaporated metal to form a continuous film across both the wafer surface and the photoresist edges, making lift-off unreliable and often leaving metallic "fences" around the contact perimeter. The dual-layer stack mitigates this by exploiting the higher developer solubility of the LOR base layer. During development, the LOR recedes laterally more than the S1803 above it, producing a distinct undercut profile. This physical overhang breaks the continuity of the deposited metal film, ensuring that the metal on top of the photoresist is electrically and mechanically separated from the device contacts. Consequently, solvent-based lift-off yields sharply defined openings free of edge defects.

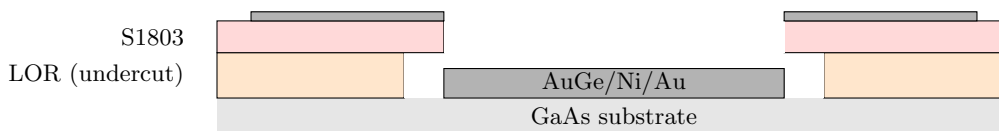


Figure 3.3: Dual-layer LOR/S1803 resist stack after development showing the wider LOR opening that creates an undercut for clean metal lift-off.

Following lithography, the Ohmic contacts were formed by thermally evaporating a AuGeNi eutectic pellet. This alloy is commonly used for n-type GaAs Ohmic contacts because each element plays a specific role during annealing. Germanium diffuses into the GaAs and acts as a donor, creating a highly doped region beneath the contact. Nickel improves the uniformity and adhesion of the metal film and supports the diffusion process, whilst gold provides a low-resistance conduction path between the contact pad and the semiconductor.

To activate the contacts, the sample underwent Rapid Thermal Annealing (RTA). It is during this thermal cycle that the transition from rectifying to Ohmic behaviour occurs. The heat drives the Germanium atoms to diffuse through the AlGaAs layers and into the GaAs lattice, locally creating the n^+ doped region. This effectively "thins" the potential barrier at the interface, establishing a low-resistance tunnelling path that connects the drain and source electrically to the 2DEG.



Figure 3.4: Ohmic contacts under x5 microscopy following lift-off.

3.4 Gate Formation and Final Inspection

With the Ohmic contacts established and annealed, a third photolithography step was performed to define the Schottky gate region. As with the previous stage, a dual-layer resist stack was utilised to ensure a clean lift-off. The gate electrode was aligned centrally between the source and drain contacts; its placement is critical, as it provides the electrostatic control required to modulate the carrier density within the channel.

Following development, a Ti/Au stack was evaporated to form the gate electrode. In this configuration, the Titanium (Ti) layer serves as a vital adhesion layer to the semiconductor surface, while the thick Gold (Au) layer provides a stable Schottky barrier and minimises gate resistance. Unlike the Ohmic contacts, the gate metallisation is not annealed; preserving the Schottky barrier height is essential for minimising leakage current and maintaining effective depletion control over the channel.

After metallisation, the resist was removed using solvent lift-off to reveal the final device structure. A small-number of lift-off defects were observed under the microscope (Figure 3.5) and subsequently removed using brief, gentle ultrasonic cleaning. The removal of these artefacts is essential for device reliability: they can otherwise cause electrical short circuits, particularly around the sensitive gate edges.

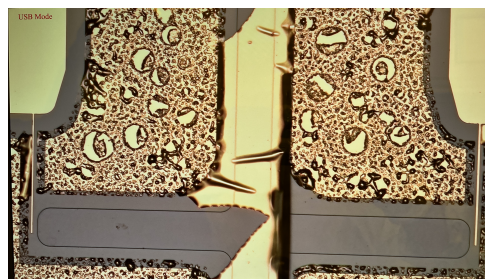


Figure 3.5: Lift-off defects (circled) observed on completed devices prior to ultrasonic cleaning.

The completed devices were examined using optical microscopy to ensure process integrity. The mesas exhibited clean, well-defined sidewalls; the Ohmic pads were properly positioned; and the gate electrodes showed sharp, uninterrupted edges with no evidence of bridging or residual flakes. Overall, the fabrication process yielded a high-quality array of depletion-mode HEMTs suitable for subsequent electrical characterisation.

3.5 Summary of Fabricated Device Set

The fabrication process yielded a structured array of depletion-mode GaAs/AlGaAs HEMT devices arranged in a grid across the wafer. This enabled multiple device geometries to be fabricated simultaneously under identical process conditions, facilitating systematic electrical characterisation while minimising process-induced variation.

In total, fifteen HEMT devices were fabricated and subsequently characterised. The array comprises five distinct columns, each containing three nominally identical devices. Each column corresponds to a specific lateral device geometry defined by a combination of gate length and source–drain separation, outlined in Table 3.1.

Table 3.1: Summary of fabricated HEMT device geometries by column

Parameter	Col 1	Col 2	Col 3	Col 4	Col 5
Gate length (μm)	10	8	6	4	4
Ohmic separation (μm)	40	32	28	24	20

Across the array, the gate length was reduced from $10\mu\text{m}$ to $4\mu\text{m}$. In parallel, the lateral separation between the ohmic contacts was also scaled between columns. The quoted source–drain separation refers to the combined distance between ohmic contacts in the double-gate layout; for example, a separation of $40\mu\text{m}$ corresponds to two $20\mu\text{m}$ access regions between the source and each drain, with the gate located centrally between the contacts. Demonstrated in Figure 3.6.

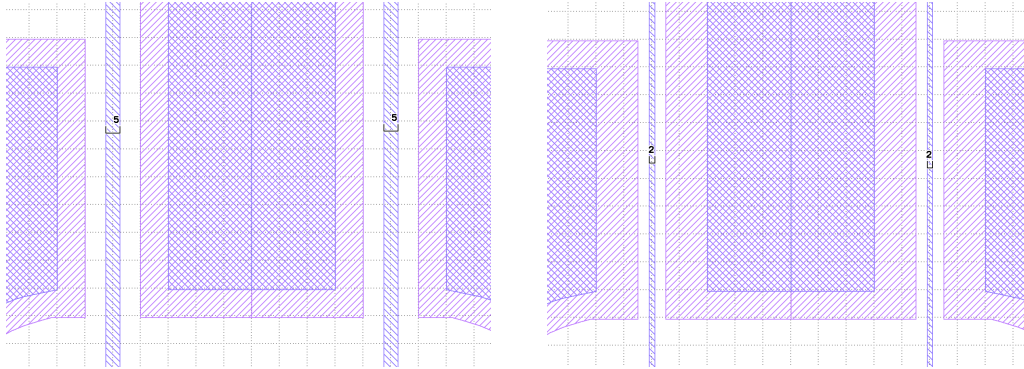


Figure 3.6: Mask layouts illustrating the extremes of the fabricated device geometries: gate length $10\mu\text{m}$ with ohmic separation $40\mu\text{m}$ (left) and gate length $4\mu\text{m}$ with ohmic separation $20\mu\text{m}$ (right).

Notably, the final two columns share the same nominal gate length of $4\mu\text{m}$ but differ in source–drain separation. This design choice allows the influence of lateral access resistance and transport effects to be assessed independently of gate-controlled electrostatics during electrical analysis.

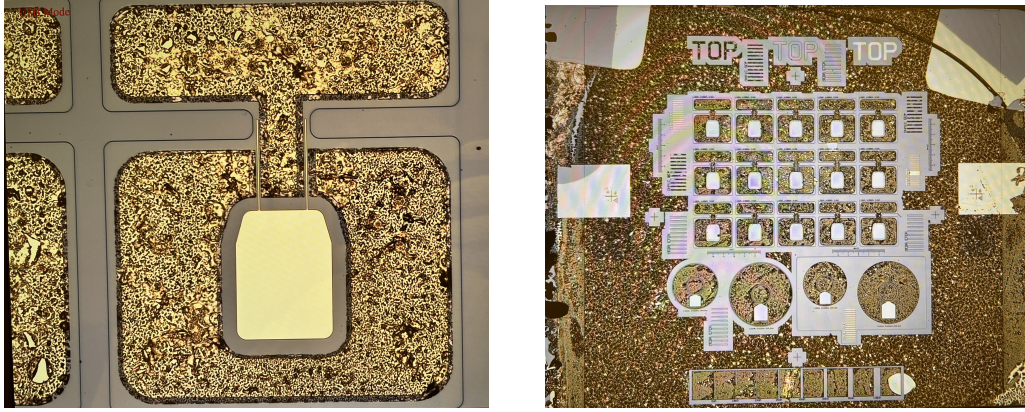


Figure 3.7: Final fabricated depletion-mode GaAs/AlGaAs HEMT device (left) and array of fabricated devices across the wafer (right).

The three nominally identical devices within each column provide a direct measure of device-to-device variability and fabrication repeatability for a given geometry. This device set therefore establishes a well-controlled baseline. The electrical characterisation presented in Chapter 4 uses this baseline to assess depletion-mode device behaviour and to validate the unrecessed fabrication process as a reference for subsequent enhancement-mode development via controlled gate recession.

CHAPTER 4

Baseline Electrical Characterisation

Here we will collate all unrecessed device measurements, focusing on the metrics that will later be compared against recessed devices. Figures and statistics from the interim report can be updated with any refined analysis.

4.1 Measurement Setup

Describe the B1500A configuration, probe station workflow, temperature controls, and data-processing scripts.

4.2 Output Characteristics

Summarise representative ID–VDS curves, contact resistance extraction, and implications for process quality.

4.3 Transfer Characteristics and Threshold Extraction

Document VT extraction methods, gm behaviour, and variability across device geometries.

4.4 Gate Leakage Behaviour

Provide leakage trends, discuss interface quality, and establish the reference point for recessed devices.

CHAPTER 5

Gate Recess Process Engineering

This chapter will describe the Semester 2 process innovations that enabled precise gate recessing, including calibration structures, etch timing models, and oxide management.

5.1 Process Objectives

Define the targeted recess depths, allowable tolerances, and link to the threshold-voltage goals set in Chapter 1.

5.2 Etch Chemistry Selection and Calibration

Report measured etch rates, agitation schemes, and profilometry data for the $\text{H}_2\text{SO}_4:\text{H}_2\text{O}_2:\text{H}_2\text{O}$ solution. Include lessons from calibration chips.

5.3 Native Oxide Mitigation

Detail the HCl dip procedure, timing constraints, and any surface-analysis results.

5.4 Mask and Layout Strategy

Explain the quadrant-based recess matrix, sample labelling convention, and any cleanroom logistics considerations.

CHAPTER 6

Enhancement-Mode Iteration 1: Fabrication and Results

6.1 Fabrication Process

6.2 Electrical Results

6.3 Analysis and Diagnosis

CHAPTER 7

Enhancement-Mode Iteration 2: Revised Mask and Results

7.1 Fabrication Process

7.2 Electrical Results

7.3 Analysis and Diagnosis

CHAPTER 8

Enhancement-Mode Iteration 3: Refined Process and Results

8.1 Fabrication Process

8.2 Electrical Results

8.3 Analysis

CHAPTER 9

Discussion and Failure Analysis

9.1 Summary of Experimental Outcomes

9.2 Failure Analysis

9.3 Process Limitations

9.4 Project Management Reflection

CHAPTER 10

Conclusions and Future Work

10.1 Conclusions

10.2 Future Work

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APPENDIX A

Supplementary Data and Process Logs

This appendix will gather supporting material such as extended etch-rate measurements, cleanroom log excerpts, supplementary microscope images, and code listings referenced throughout the report. Populate the subsections below as data becomes available.

A.1 Etch Calibration Tables

A.2 Cleanroom Session Notes

A.3 Additional Figures

A.4 Data Processing Scripts